

U.S. IT Hardware

Memory & Storage: What we learned from our NDR with STX Management and recent NAND & HDD earnings



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Seagate has strong multi-year demand visibility, with expected volumes initially established through LTAs and then converted into firm PO (purchase orders) 4-5 quarters before shipment. Therefore, the June quarter's ~6% QoQ pricing increase primarily reflected pricing negotiated roughly four quarters earlier. Seagate also keeps a small amount of capacity uncommitted, allowing it to capture higher spot-market pricing on incremental volume. **Management expects HDD ASP growth to continue and potentially further accelerate as prior HAMR introductory discounts roll off, while edge and IoT products remain more flexible and can see faster price increases than nearline products.**

Industry HDD supply remains highly disciplined, as Seagate, WDC, and Toshiba are not adding meaningful unit capacity. Instead, Seagate expects to drive roughly 25% annual nearline exabyte growth through HAMR-based increases in capacity per drive, despite HAMR requiring more manufacturing resources and factory space. As the only vendor aggressively scaling HAMR, we believe STX is positioned to gain exabyte share while constrained industry supply supports favorable market dynamics.

Management sees HDD demand being driven by cloud, enterprise, AI training, and increasingly AI inference workloads. **The key takeaway is that AI creates more data, and most of that data ultimately ends up on low-cost HDD storage.** Longer data retention periods, combined with demand that is primarily for net-new capacity rather than refreshes, remain strong structural tailwinds for HDDs.

Management views neoclouds as a potential long-term HDD demand opportunity. While most neocloud providers currently rely on hyperscalers for storage, future expansion into full infrastructure platforms could require their own storage deployments, creating an additional source of HDD demand beyond traditional cloud customers.

Earnings takeaways: We believe the HDD pricing cycle is in its early innings with a long runway for continued ASP growth. Pricing only began inflecting in the March quarter and has been increasing steadily, supported by strong demand and disciplined industry supply. **STX is our top pick due to its HAMR leadership.** Its lead in areal density scaling should drive faster exabyte shipment growth, better cost reductions, and greater share gains versus peers. While **WDC** also posted a strong quarter and continues to benefit from improving industry pricing, its slower HAMR transition limits its cost decline relative to STX. **As a result, we remain constructive on WDC, but continue to view STX as the best-positioned beneficiary of the HDD upcycle. We also maintain Outperform rating on SNDK** and still see significant upside particularly as 2/3 of volumes on LTAs with 80% floor gross margin should help improve earnings stability.

STX is our top pick within the sector, due to continued HDD pricing strength, supply discipline, and HAMR leadership.

BERNSTEIN TICKER TABLE

Ticker	Rating	6 Aug 2026			TTM Rel. Perf.	Adjusted EPS				Adjusted P/E (x)		
		Cur	Closing Price	Price Target		Cur	2026A	2027E	2028E	2026A	2027E	2028E
STX (Seagate)	O	USD	852.95	1,350.00	457.7%	USD	15.53	40.61	64.40	54.9	21.0	13.2
WDC (Western Digital)	O	USD	451.52	770.00	490.5%	USD	10.19	23.39	36.58	44.3	19.3	12.3
SNDK (SanDisk)	O	USD	1,258.58	3,000.00	2868.0%	USD	70.08	238.36	272.09	18.0	5.3	4.6
SPX			7,709.96									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We rate SNDK Outperform with TP of \$3000. We rate STX Outperform with TP of \$1350. We rate WDC Outperform with TP of \$770.

DETAILS

Seagate uses two main forms of customer commitments: LTAs and POs. LTAs are multi-year agreements that primarily specify expected volumes, but generally do not lock in pricing or detailed product mix beyond the near-term period. **These LTA volumes are then converted into firm PO (purchase orders) approximately 4-5 quarters before shipment.** Each purchase order specifies volume, ASP, delivery timing, number of drives, and capacity per drive, and is relatively rigid once signed. Given approximately 3 quarters of production lead time, Seagate has enough time to set firm purchasing orders before committing manufacturing resources, providing strong visibility. The vast majority of Seagate's expected CY2027 exabyte shipments are already covered by purchase orders.

Nearline data center pricing is largely locked for the next 4-5 quarters through firm purchase orders, meaning reported pricing generally reflects negotiations completed several quarters earlier. Management indicated that the 6% qoq pricing increase realized in the June quarter was primarily based on pricing agreed roughly four quarters ago, with some additional benefit from market-priced incremental exabytes not under order. Seagate intentionally commits slightly less than its maximum production capability, allowing incremental in-quarter output to be sold at substantially higher market prices and providing a mechanism to test underlying demand. The pricing negotiations are staggered; therefore, STX enjoys ASP QoQ increases each quarter.

Non-nearline edge and IoT products generally do not have firm purchase orders or LTAs, allowing Seagate to raise prices more quickly. Data center and edge margins are currently not materially different, although data center is growing faster and edge demand is more seasonal as the consumer business is stronger in Sep-Dec quarters. Management expects steady HDD ASP growth to continue at a pace similar to, or potentially better than, recent quarters as prior introductory discounts on new HAMR products roll off.

None of STX, WDC, or Toshiba are adding drive unit capacity, keeping industry-wide HDD supply structurally disciplined. Seagate management explained that its factories are effectively full, and rather than expanding units, management is targeting roughly 25% CAGR in nearline exabyte shipments purely by increasing capacity per drive via HAMR. In fact, the HAMR transition itself requires longer production cycle times, more equipment, and additional factory space just to sustain current unit output, leaving little practical room for unit growth — and building a new factory would take at least two years, which management has no plans to do. Since STX is the only player seriously ramping HAMR, it also holds the best areal density potential in the industry, a scaling advantage neither WDC nor Toshiba can currently match. As a result, Seagate is directing available exabytes primarily to existing CSP customers and expects to grow exabyte share even with unit share unchanged, while the broader industry's inability to scale supply keeps supply disciplined and positions STX as the clear share gainer as the cycle progresses.

Management was unequivocal that HAMR remains the primary technology enabling future HDD capacity growth and economics. Rather than expanding drive unit production, Seagate intends to grow exabytes shipped through higher-capacity drives. The company argued that adding platters and heads offers diminishing returns, with moving from 10 to 11 disks increasing capacity by only ~10%, whereas migrating from 30TB to 40TB drives provides roughly a 30% capacity uplift. While HAMR requires more equipment, longer manufacturing cycle times, and additional factory space than ePMR, the company remains focused on technology transitions rather than factory expansion. Management stated they are effectively done adding manufacturing capacity and instead expect to achieve roughly 25% annual nearline exabyte growth through HAMR adoption, with the nearline portfolio already at approximately 40% HAMR mix and expected to become predominantly HAMR over time. Customer demand is increasingly concentrated on higher-capacity drives, and Seagate believes the current supply environment remains constrained despite factories operating at full utilization.

Management emphasized that HDD demand is not being driven by a single AI use case but rather by a combination of traditional cloud growth, enterprise storage, training workloads, and AI inference. The company believes it is increasingly difficult to separate what portion of demand is "AI" versus traditional storage demand because both ultimately result in growing data creation. The most important AI-related takeaway was that inference may become a more meaningful HDD demand driver than training over time. Every inference interaction generates additional context, outputs, metadata, and content that must be retained and stored, supporting longer-term demand for mass-capacity storage. Management also highlighted that data retention periods continue to lengthen, creating a structural tailwind for HDDs.

While NAND and SSDs are essential for high-performance workloads, KV cache, and warm storage, management noted that warm storage represents only a small portion of total exabytes, whereas cold storage remains significantly larger. The underlying message was that AI increases compute activity, compute creates data, and nearly all of that data ultimately finds its

way to lower-cost HDD storage. **Demand today is overwhelmingly for new capacity rather than refreshes**, as customers would rather add incremental drives than retire existing drives and lose revenue-generating storage capacity. HDDs are typically warranted for five years but often remain in service for six to seven years.

Neoclouds could become an incremental source of HDD demand over time. Today, most neocloud providers offer compute while relying on CSPs for storage, meaning Seagate's HDD shipments remain concentrated with hyperscalers. However, if neoclouds evolve into broader infrastructure platforms and seek greater independence from CSPs, they will likely need to build their own storage footprints, creating an additional layer of HDD demand. While this opportunity remains early, as neoclouds are not yet HAMR-qualified customers and pricing discussions have not begun, management views their emergence as a potential long-term growth vector beyond traditional cloud demand.

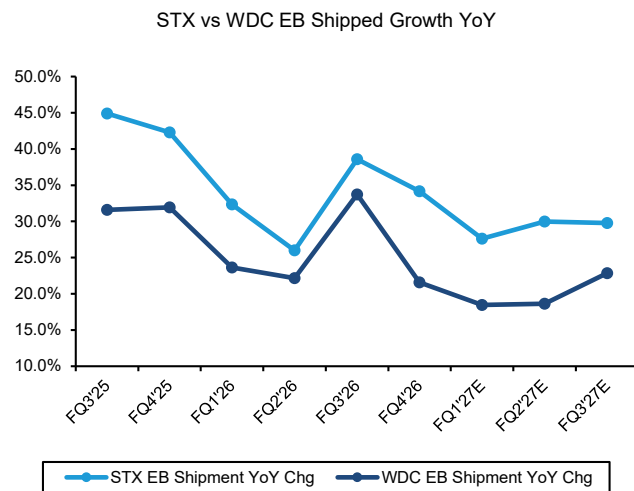
Earnings takeaways:

SNKD delivered a great FQ4'26. While some may worry its growing LTA coverage **caps some near-term upside** and is slowing ASP growth, **we believe it is important to note that greater LTA coverage is positive** for the stock's multiple and rerating potential: LTAs provide robust downside protection, **make nosebleed earnings more sustainable, and should reduce perceived cyclicality.** For additional details, please see [Sandisk FQ4'26: As LTAs hit 2/3 of volume, ASP increases slow but more downside protection than previously expected.](#)

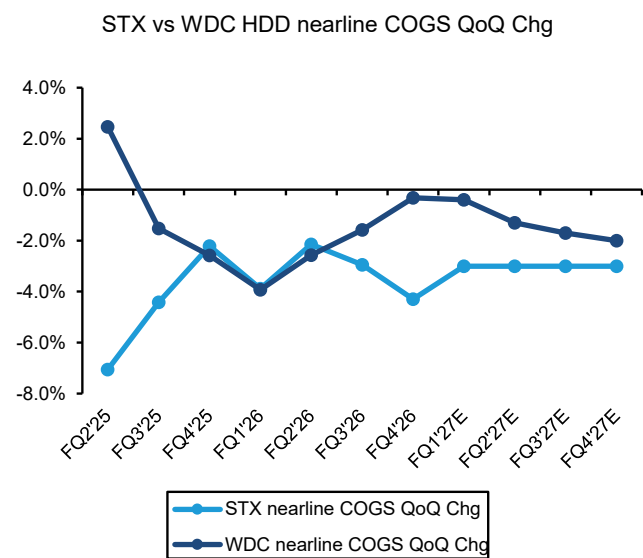
Why the HDD cycle has further to run: We believe the HDD ASP upcycle is just getting started. Pricing only inflected roughly two quarters ago and has since risen steadily in the mid-to-high single-digit range qoq— an early and steady ramp compared to memory's steep spike. Looking ahead, we would not be surprised if HDD qoq pricing growth actually steps up a bit from recent quarters, given strong end-market demand. Net, we see a long and steady runway ahead for HDD pricing.

STX vs. WDC: Within that favorable HDD backdrop, HAMR leadership is the key differentiator. Its lead in HAMR-driven areal density scaling supports faster exabyte shipment growth and deeper cost declines than peers. Moreover, because transitioning from ePMR to HAMR manufacturing takes meaningfully longer than moving between successive HAMR generations (e.g., Mozaic 3 to Mozaic 4), WDC's HAMR ramp could prove slower than expected, whereas Seagate's head start lets it iterate to higher-capacity HAMR generations more quickly. **Combined with disciplined industry supply and the ASP upcycle, this makes STX a steady long-term compounder, and we reiterate it as the top pick in our coverage.** For additional details, please see [STX FQ4'26: This ain't over... Raising estimates and TP to \\$1350; Reiterate Outperform, Top Long-term pick.](#)

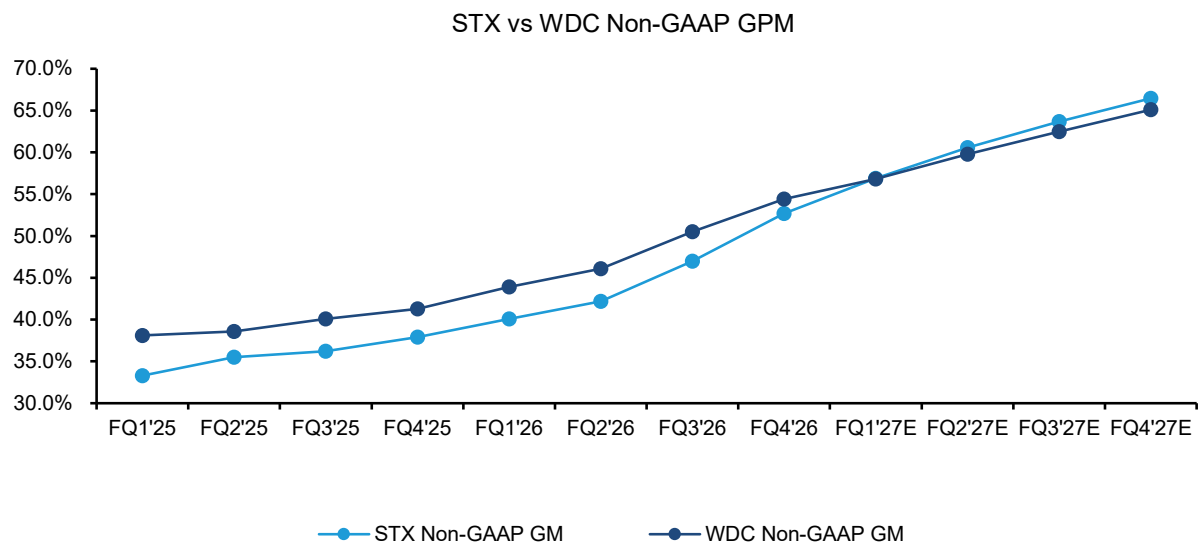
On the other hand, WDC also delivered a strong beat-and-raise quarter, driven largely by favorable pricing, but its guidance falls short of STX's because WDC is lagging on HAMR. That lag constrains WDC's areal density scaling, which in turn limits its exabyte shipment growth and cost-reduction trajectory relative to STX. **A rising tide is lifting all boats, and we remain constructive on both names — but STX's HAMR lead is why it is our top pick.** For additional details on WDC, please see: [WDC FQ4'26: Waiting for HAMR time, but pricing saves the day; Reiterate Outperform, TP raised to \\$770.](#)

EXHIBIT 1: STX shipment YoY Growth above WDC's due to superior areal density scaling


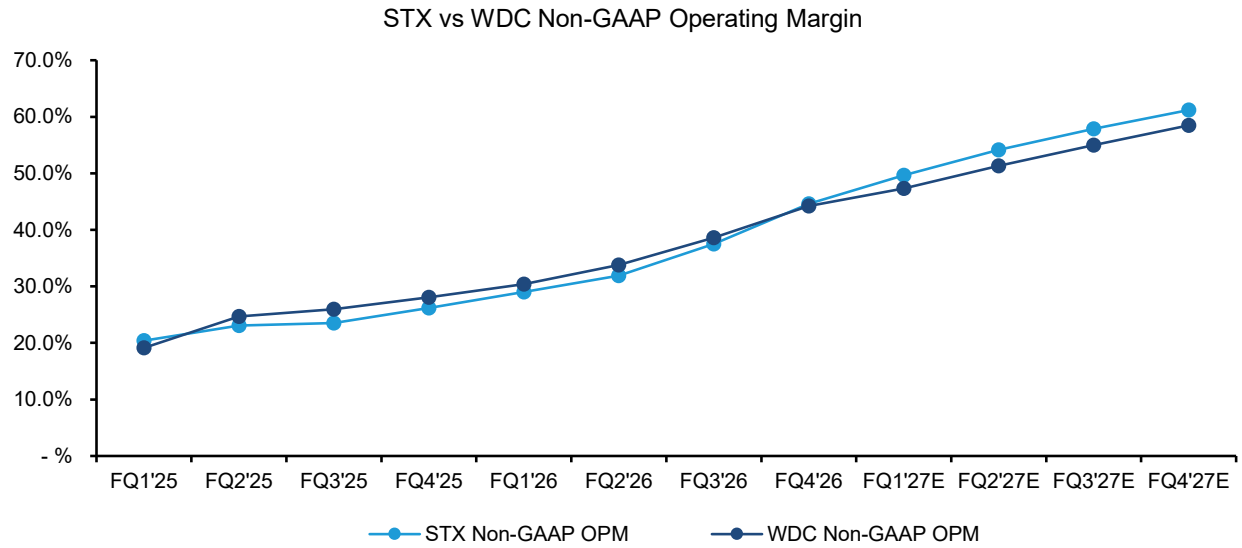
Source: Company reports, Bernstein estimates and analysis

EXHIBIT 2: STX nearline COGS decline is larger than WDC's due to STX's HAMR leadership


Source: Company reports, Bernstein estimates and analysis

EXHIBIT 3: STX will enjoy superior GPM as the leader in HAMR


Source: Company reports, Bernstein estimates and analysis

EXHIBIT 4: **STX will enjoy better OPM due to HAMR leadership**

Source: Company reports, Bernstein estimates and analysis

EXHIBIT 5: **The four stages of AI**

	Stage 1: Training <i>Pre-training</i>	Stage 2: Basic chatbot <i>inference</i>	Stage 3: Advanced inference <i>Long-context, RAG, reasoning</i>	Stage 4: Agentic AI <i>Tool use, memory, workflows</i>	
HBM (DRAM)	Extreme	High	Extreme	High	
Conventional DRAM (DDR5)	High	Moderate	High	Extreme	
NAND Flash / Enterprise SSD	High	Moderate	Extreme	Extreme	<i>SNDK</i>
HDD (Nearline)	High	Low/Indirect	Moderate	High	<i>STX</i>
Tape (LTO / archive)	Moderate	Low	Low	Moderate	

Legend:
 Extreme
 High
 Moderate
 Low/Indirect

Source: Bernstein analysis

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VALUATION METHODOLOGY**Seagate Technology PLC**

We value Seagate at 21x our FY28 EPS of \$64.40, implying \$1,350 per share. We believe Seagate's improving fundamentals, >70% 5year EPS CAGR, and HAMR leadership easily justifies a 21x P/E, and arguably much higher.

Western Digital Corp

We rate WDC Outperform and have a TP of \$770, on 21x our FY28 EPS.

SanDisk Corp

We value SNDK at \$3000 on 11x FY28 EPS, or 14x through cycle EPS (FY26-30 avg EPS).

RISKS**Seagate Technology PLC**

The biggest risks to the downside on Seagate and to our price target are that: The HDD market is heavily exposed to hyperscale cloud, and could be impacted by a cloud capex digestion or else incremental pressure from changes in hyperscaler buying patterns. 2) WDC could close the gap in HAMR more quickly than expected, shrinking the scope for market share gain and margin improvement. 3) NAND could see better than expected technology improvement and take share from HDDs.

Western Digital Corp

The biggest risks to the downside on WDC and to our price target are that: 1) The HDD market is heavily exposed to hyperscale cloud, and could be impacted by a cloud capex digestion or else incremental pressure from changes in hyperscaler buying patterns; 2) NAND could see better than expected technology improvement and take share from HDDs; and 3) WDC could face significant risks ahead with its HAMR technology transition which could provide a drag on gross margin and EPS.

SanDisk Corp

The biggest risks to the downside on Sandisk and to our price target are that: 1) Near-term numbers look high and face cyclical downside from NAND; 2) SNDK's disclosures and broader investor communications have been confusing, potentially dissuading more quality-focused investors; and 3) NAND weakness could extend beyond the current cycle and be more structural, in which case Sandisk's DCF value could be structurally lower and the impaired value of its assets may be materially below replacement cost.

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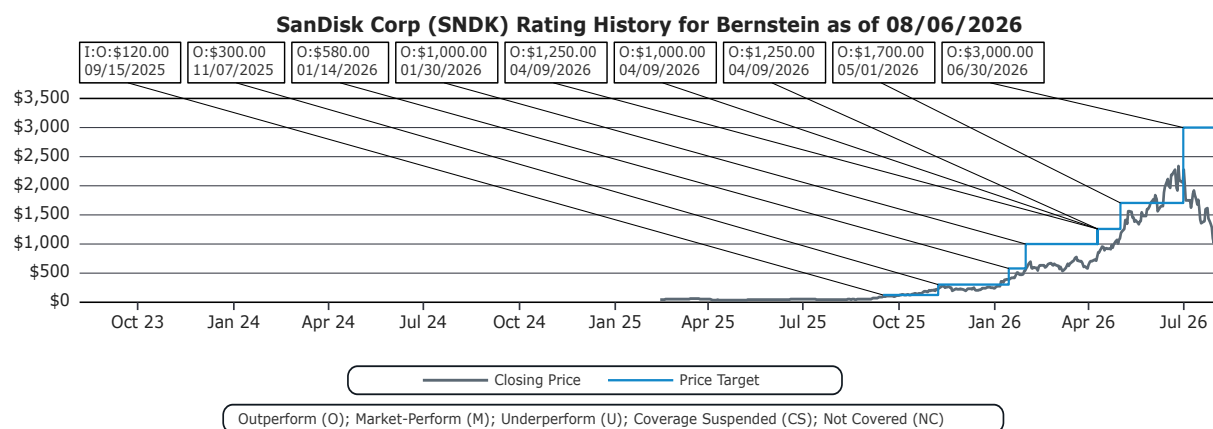
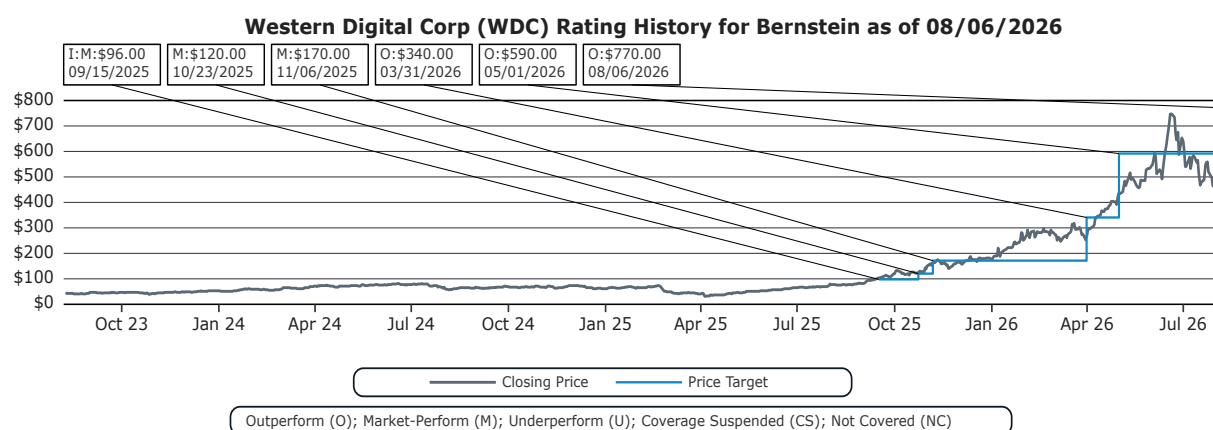
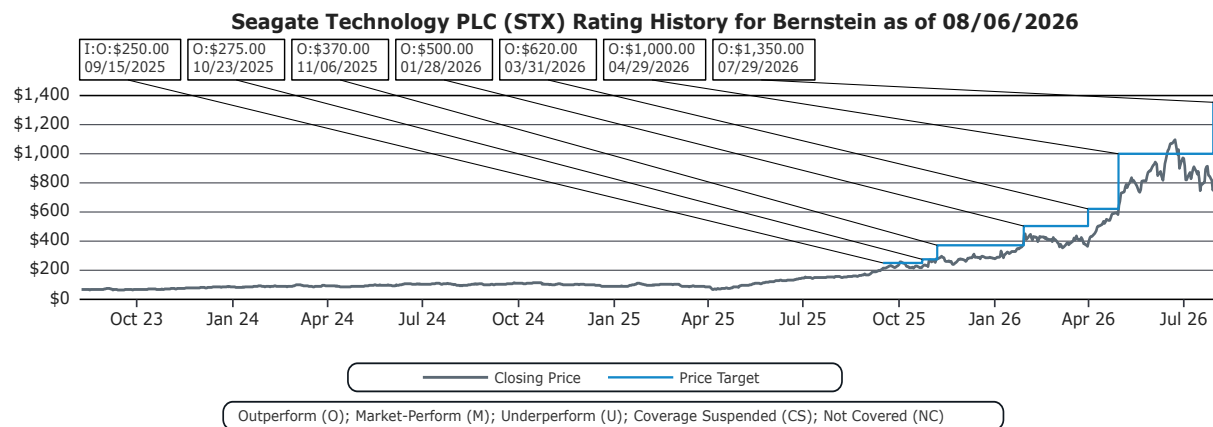
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Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
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